Chapter 4

Vector geometry



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4.1 Vectors and Lines

Vertor in $\mathbf{R}^3$

As in the plane, we introduce vectors by identifying each point $P(x, y, z)$ with the vector

$$v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

in $\mathbf{R}^3$, represented by the **arrow** from the origin to P as in Figure 1

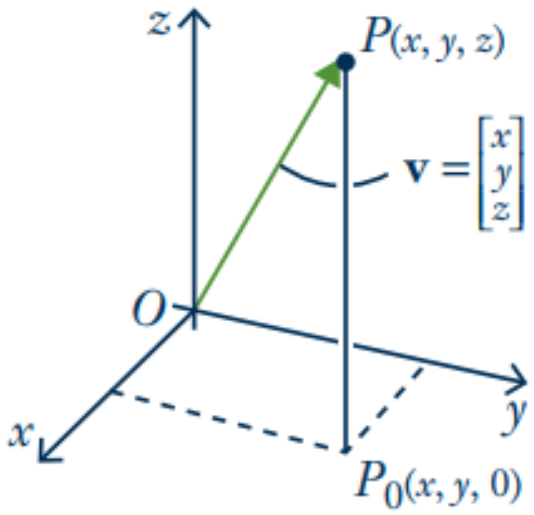


FIGURE 1

Length and Direction

Theorem 1

Let $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be a vector.

(1) $\|v\| = \sqrt{x^2 + y^2 + z^2}$

(2) $v = 0$ if and only if $||v|| = 0$

(3) $||av|| = |a|\ ||v||$ for all scalars a.

Length and Direction

EXAMPLE 1

If $v = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ then $\|v\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$.

Similarly if $v = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$ in 2-space then $\|v\| = \sqrt{3^2 + (-4)^2} = 5$

Length and Direction

Theorem 2

- *Let* $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$ *be vectors in* R^3. *Then* $\mathbf{v} = \mathbf{w}$ *as matrices if and only if* $\mathbf{v}$ *and* $\mathbf{w}$ *have the same direction and the same length.*

Geometric Vectors

Definition 4.1

Suppose that A and B are any two points in R^3. *In Figure 4 the line segment from A to B is denoted* $\overrightarrow{AB}$ *and is called the* **geometric vector** *from A to B. Point A is called the* **tail** *of* $\overrightarrow{AB}$, *B is called the* **tip** *of* $\overrightarrow{AB}$, *and the* **length** *of* $\overrightarrow{AB}$ *is denoted* $\left\|\overrightarrow{AB}\right\|$.

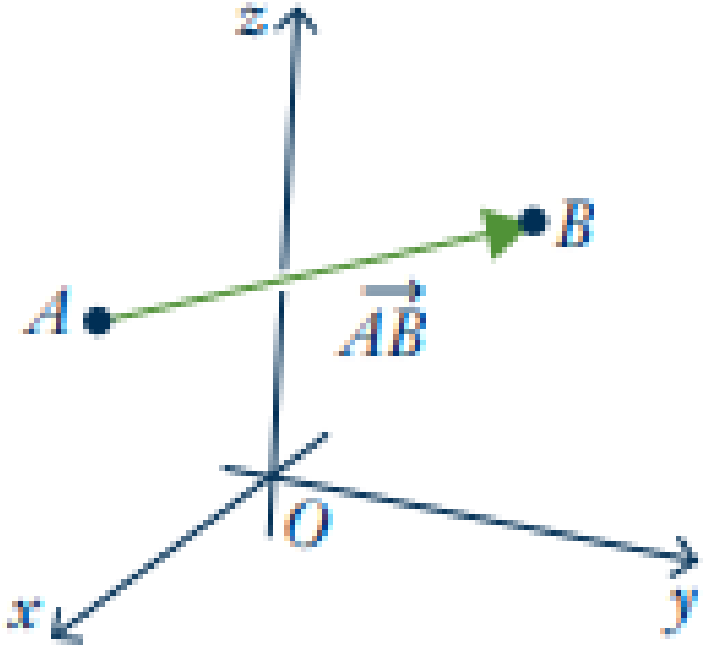


FIGURE 4

The Parallelogram Law

- **The Parallelogram Law**

In the parallelogram determined by two vectors **v** *and* **w**, *the vector* **v** + **w** *is the diagonal with the same tail as* **v** *and* **w**.

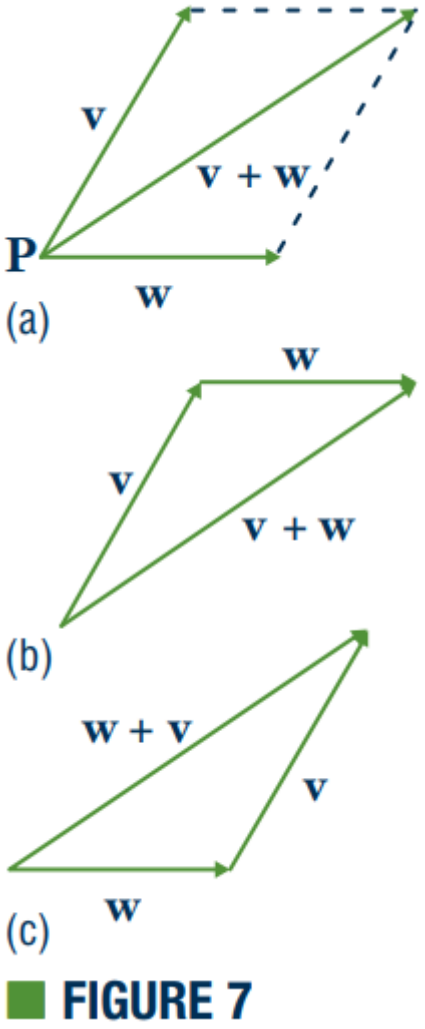
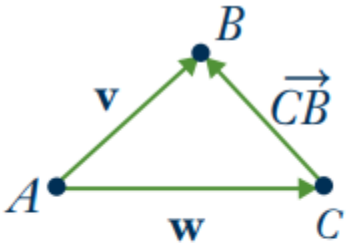


FIGURE 7

Theorem 3

If **v** *and* **w** *have a common tail, then* **v** - **w** *is the vector from the tip of* **w** *to the tip of* **v**.



Theorem 4

Let $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ be two points. Then:

1. $\overrightarrow{P_1P_2} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix}$

2. The distance between P_1 and P_2 is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

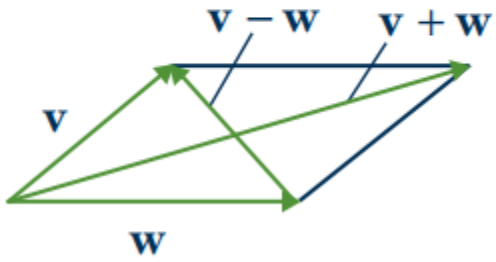


FIGURE 9

EXAMPLE 3

The vector from $P_1(2, -1, 3)$ to $P_2(1, 1, 4)$ is

$$\overrightarrow{P_1P_2} = (1-2, 1-(-1), 4-3) = (-1, 2, 1)$$

and the distance between P_1 and P_2 is

$$\sqrt{(-1)^2 + 2^2 + 1^2} = \sqrt{6}$$

Scalar Multiplication

If a is a real number and $v \neq 0$ is a vector then:

(1) The length of av is $||av|| = |a|\ ||v||$.

(2) If $av \neq 0$, the direction of av is

$$\begin{cases} \text{the same as } v \text{ if } a > 0, \\ \text{oppostite to } v \text{ if } a < 0. \end{cases}$$

Scalar Multiplication

A vector **u** is called a **unit vector** if ‖**u**‖ = 1. Then

$$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{and } k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

are unit vectors, called the **coordinate** vectors.

EXAMPLE 4

If **v** ≠ **0** then $\frac{1}{\|v\|} v$ is the unique unit vector in the same direction as **v**.

Definition 4.2

Two nonzero vectors are called **parallel** *if they have the same or opposite direction.*

Theorem 5

Two nonzero vectors v and w are parallel if and only if one is a scalar multiple of the other.

EXAMPLE 7

Given points $P(2, -1, 4)$, $Q(3, -1, 3)$, $A(0, 2, 1)$, and $B(1, 3, 0)$, determine if $\overrightarrow{PQ}$ and $\overrightarrow{AB}$ are parallel.

Lines in Space

We call a nonzero vector $\mathbf{d} \neq \mathbf{0}$ *a* **direction vector** *for the line if it is parallel to* $\overrightarrow{AB}$ *for some pair of distinct points A and B on the line.*

Vector Equation of a Line

The line parallel to $\mathbf{d} \neq \mathbf{0}$ *through the point with vector* $\mathbf{p}_0$ *is given by*

$$\mathbf{p} = \mathbf{p}_0 + t\mathbf{d}, \qquad t \text{ any scalar}$$

In other words, the point $\mathbf{p}$ *is on this line if and only if a real number* t *exists such that* $\mathbf{p} = \mathbf{p}_0 + t\mathbf{d}$.

In component form the vector equation becomes

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Equating components gives a different description of the line.

Parametric Equations of a Line

The line through $P_0(x_0, y_0, z_0)$ *with direction vector* $d = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$ *is given by*

$$x = x_0 + ta$$
$$y = y_0 + tb \qquad t \text{ any scalar}$$
$$z = z_0 + tc$$

In other words, the point $P(x, y, z)$ *is on this line if and only if a real number* t *exists such that* $x = x_0 + ta$, $y = y_0 + tb$, *and* $z = z_0 + tc$.

Examples 8

- Find the equations of the line through the points $P_0(2, 0, 1)$ and $P_1(4, -1, 1)$.

Examples 9

Find the equations of the line through $P_0(3, -1, 2)$ parallel to the line with equations

$$x = -1 + 2t$$
$$y = 1 + t$$
$$z = -3 + 4t$$

Examples 10

Determine whether the following lines intersect and, if so, find the point of intersection.

$$\begin{cases} x = 1 - 3t \\ y = 2 + 5t \\ z = 1 + t \end{cases} \qquad \begin{cases} x = -1 + s \\ y = 3 - 4s \\ z = 1 - s \end{cases}$$

Theorem 6

Pythagoras' Theorem

Given a right-angled triangle with hypotenuse c and sides a and b, then $a^2 + b^2 = c^2$

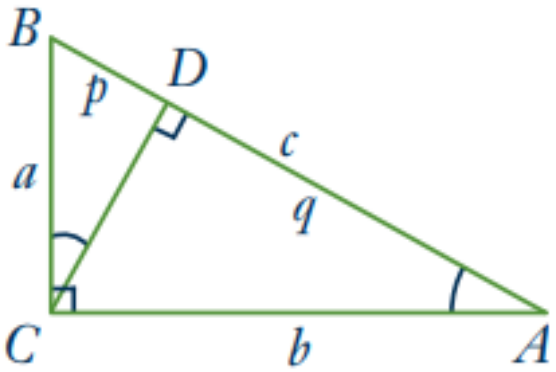
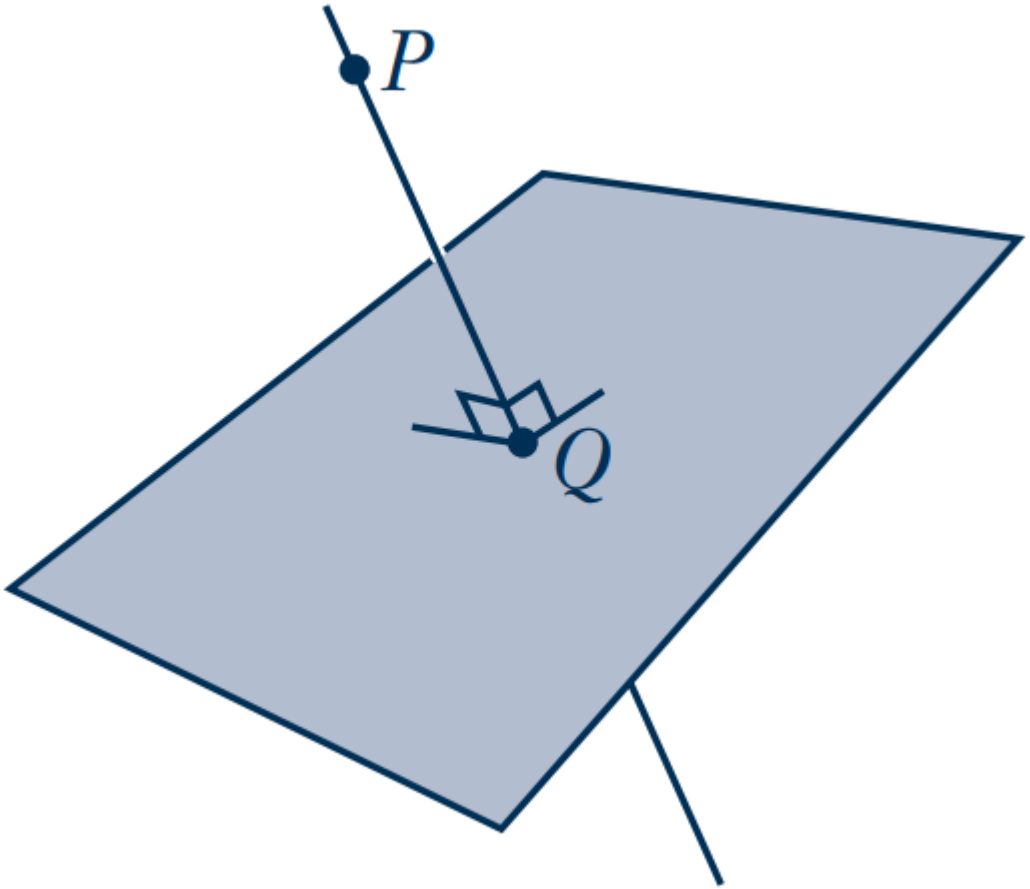


FIGURE 14

4.2. Projections and Planes



4.2. Projections and Planes

The Dot Product and Angles

Given vectors $v = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ *and* $\mathrm{w} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, *their* **dot product v · w** *is a number defined*

$$\mathbf{v} \cdot \mathbf{w} = x_1x_2 + y_1y_2 + z_1z_2 = \mathbf{v}^T\mathbf{w}$$

EXAMPLE 1

If $v = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $\mathrm{w} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$,

then $v \cdot \mathrm{w} = 2 \cdot 1 + (-1) \cdot 4 + 3 \cdot (-1) = -5$

4.2. Projections and Planes

Theorem 1

Let $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *denote vectors in* R^3 *(or* R^2*).*

1. $\mathbf{v} \cdot \mathbf{w}$ *is a real number.*
2. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$.
3. $\mathbf{v} \cdot \mathbf{0} = 0 = \mathbf{0} \cdot \mathbf{v}$.
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$.
5. $(k\mathbf{v}) \cdot \mathbf{w} = k(\mathbf{w} \cdot \mathbf{v}) = \mathbf{v} \cdot (k\mathbf{w})$ *for all scalars* k.
6. $\mathbf{u} \cdot (\mathbf{v} \pm \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} \pm \mathbf{u} \cdot \mathbf{w}$

4.2. Projections and Planes

EXAMPLE 2

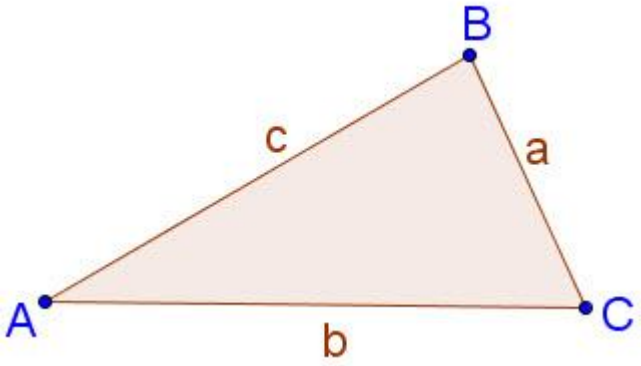
Verify that $\|\mathbf{v} - 3\mathbf{w}\|^2 = 1$ when $\|\mathbf{v}\| = 2$, $\|\mathbf{w}\| = 1$, and $\mathbf{v}\cdot\mathbf{w} = 2$.

4.2. Projections and Planes

Law of Cosines

If a triangle has sides a, b, and c then

$$c^2 = a^2 + b^2 - 2ab\cos(C).$$



Theorem 2

Let **v** *and* **w** *be nonzero vectors. If* θ *is the angle between* **v** *and* **w**, *then*

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\|\cos\theta$$

4.2. Projections and Planes

EXAMPLE 3

Compute the angle between $u = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$.

4.2. Projections and Planes

Definition 4.5

Two vectors $\mathbf{v}$ *and* $\mathbf{w}$ *are said to be* **orthogonal** *if* $\mathbf{v} = \mathbf{0}$ *or* $\mathbf{w} = \mathbf{0}$ *or the angle between them is* $\frac{\pi}{2}$.

Theorem 3

Two vectors $\mathbf{v}$ *and* $\mathbf{w}$ *are orthogonal if and only if* $\mathbf{v} \cdot \mathbf{w} = 0$.

4.2. Projections and Planes

Example 4

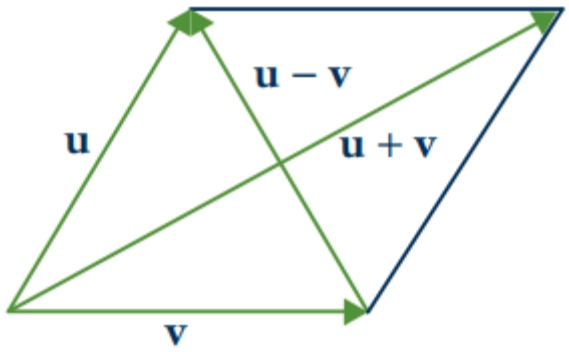
Show that the points $P(3, -1, 1)$, $Q(4, 1, 4)$, and $R(6, 0, 4)$ are the vertices of a right triangle.

4.2. Projections and Planes

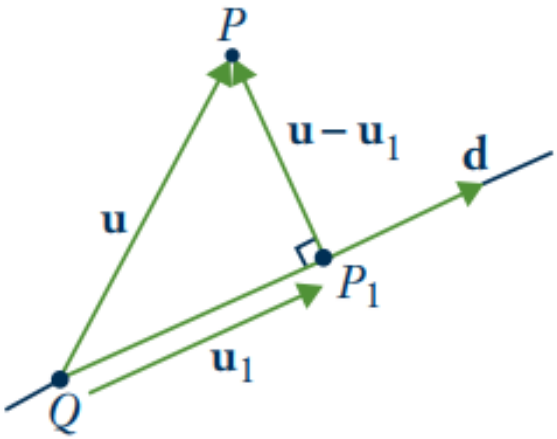
Example 5

A parallelogram with sides of equal length is called a **rhombus**. Show that the diagonals of a rhombus are perpendicular.

Solution:



4.2. Projections and Planes



If a nonzero vector **d** is specified,
then $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$, where $\mathbf{u}_1$ is parallel to **d** and
$\mathbf{u}_2 = \mathbf{u} - \mathbf{u}_1$ is orthogonal to **d**.

Suppose that **u** and $\mathbf{d} \neq \mathbf{0}$ emanate from a common tail Q. Let P be the tip of **u**, and let P_1 denote the foot of the perpendicular from P to the line through Q parallel to **d**. Then $u_1 = \overrightarrow{QP_1}$ has the required properties:

1. $\mathbf{u}_1$ *is parallel to* **d**.
2. $\mathbf{u}_2 = \mathbf{u} - \mathbf{u}_1$ *is orthogonal to* **d**.
3. $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$.

4.2. Projections and Planes

Definition 4.6

The vector $u_1 = \overrightarrow{QP_1}$ *in Figure 5 is called* **the projection of u on d.** *It is denoted*

$$\mathbf{u}_1 = \text{proj}_{\mathbf{d}}\,\mathbf{u}$$

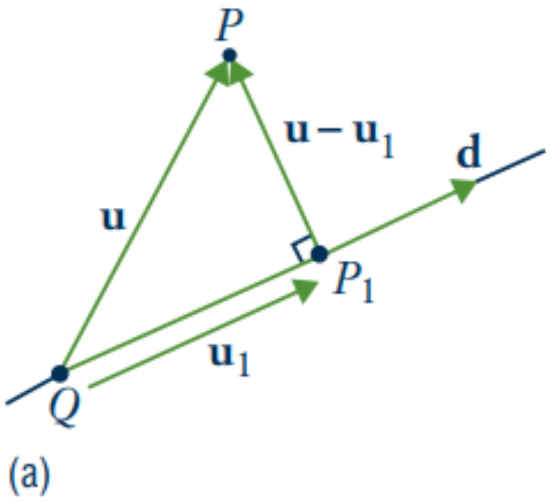


FIGURE 5

Theorem 4

Let u and $d \neq 0$ *be vectors.*

1. *The projection of* ***u*** *on* ***d*** *is given by*

$$proj_{\mathbf{d}}\boldsymbol{u} = \frac{\mathbf{u}.\mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d}$$

2. *The vector* $\mathbf{u}\text{-proj}_{\mathbf{d}}\mathbf{u}$ *is orthogonal to* **d**.

4.2. Projections and Planes

Example 7

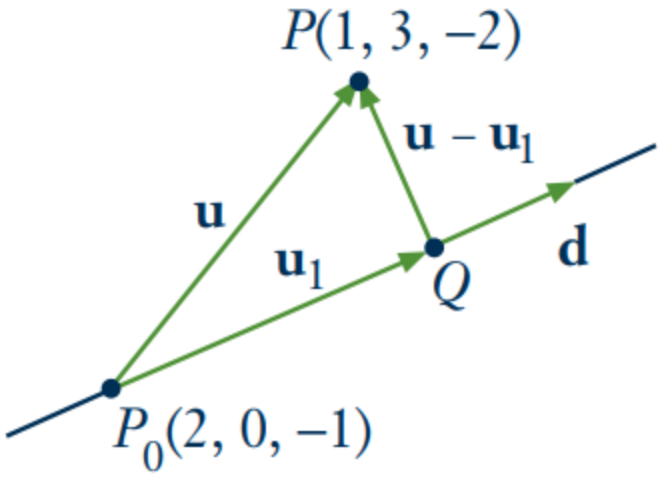
Find the projection of $\mathbf{u} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ on $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and express $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$

Where $\mathbf{u}_1$ is parallel to $\mathbf{d}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{d}$.

4.2. Projections and Planes

Example 8

Find the shortest distance (see diagram) from the point $P(1, 3, -2)$ to the line through $P_0(2, 0, -1)$ with direction vector $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$.

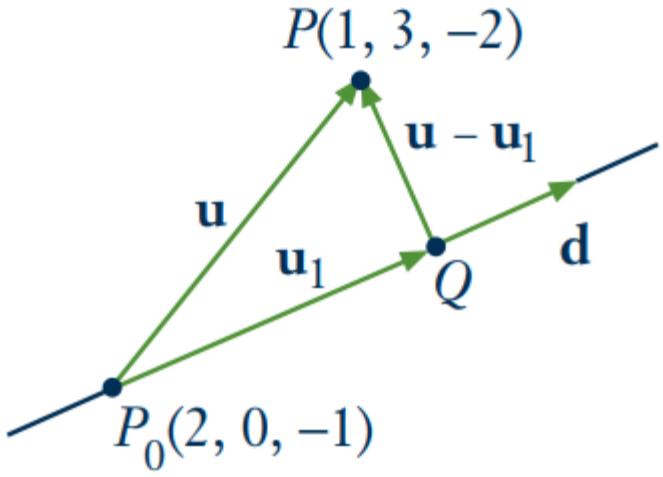


Also find the point Q that lies on the line and is closest to P.

4.2. Projections and Planes

Solution 8

$$\text{Let } u = \overrightarrow{P_0P} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}$$



and let $\mathbf{u}_1$ denote the projection of $\mathbf{u}$ on $\mathbf{d}$. Thus

$$\mathbf{u}_1 = \mathbf{proj}_{\mathbf{d}}\mathbf{u} = \frac{\mathbf{u}.\mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d} = \frac{-2-3+0}{1^2+(-1)^2+0^2}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}$$

Solution 8

We see geometrically that the point Q on the line is closest to P, so the distance is

$$\left\|\overrightarrow{QP}\right\| = \|u - u_1\| = \left\|\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}\right\| = \sqrt{3}$$

To find the coordinates of Q, let $\mathbf{p}_0$ and $\mathbf{q}$ denote the vectors of P_0 and Q, respectively. Then

$$p_0 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \text{ and } q = p_0 + u_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

Hence $Q(0, 2, -1)$ is the required point. It can be checked that the distance from Q to P is $\sqrt{3}$ as expected.

Definition 4.7

A nonzero vector **n** *is called a* **normal** *for a plane if it is orthogonal to every vector in the plane.*

Scalar Equation of a Plane

The plane through $P_0(x_0, y_0, z_0)$ *with normal* $n = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$ *as a normal vector is given by*

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

In other words, a point $P(x, y, z)$ *is on this plane if and only if* x, y, *and* z *satisfy this equation.*

EXAMPLE 9

Find an equation of the plane through $P_0(1, -1, 3)$ with $n = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ as normal.

EXAMPLE 10

Find an equation of the plane through $P_0(3, -1, 2)$ that is parallel to the plane with equation $2x - 3y = 6$.

Vector Equation of a Plane

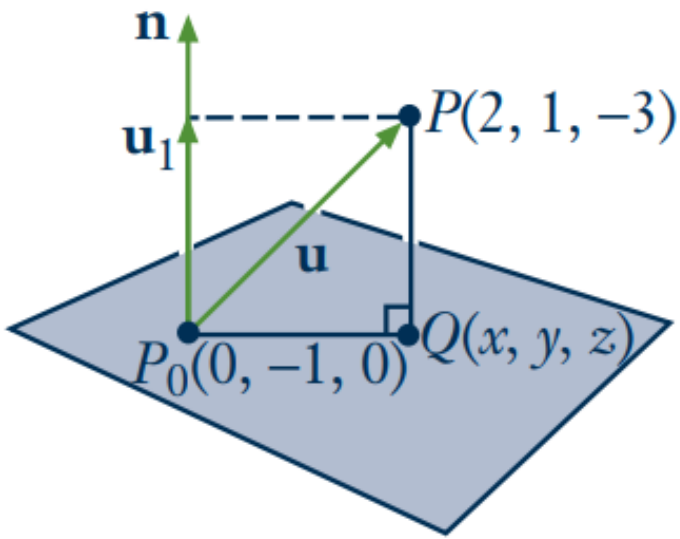
The plane with normal $\mathbf{n} \neq \mathbf{0}$ *through the point with vector* $\mathbf{p}_0$ *is given by*

$$\mathbf{n} \cdot (\mathbf{p} - \mathbf{p}_0) = 0$$

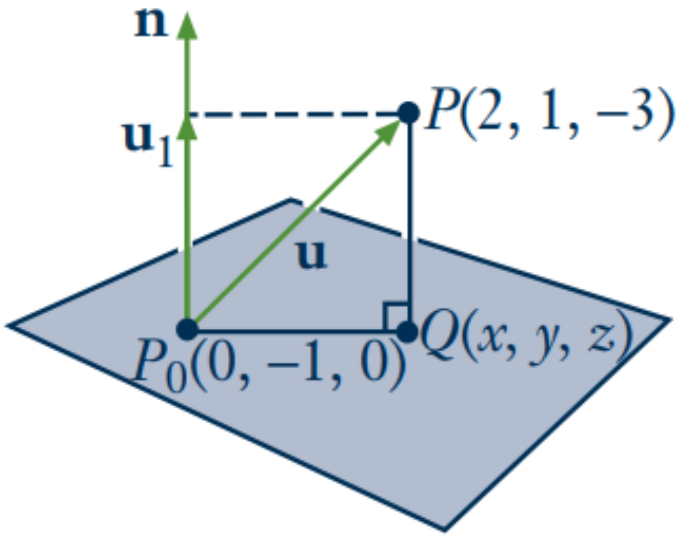
In other words, the point with vector $\mathbf{p}$ *is on the plane if and only if* $\mathbf{p}$ *satisfies this condition.*

Example 11

Find the shortest distance from the point $P(2, 1, -3)$ to the plane with equation $3x - y + 4z = 1$. Also find the point Q on this plane closest to P.



Solution 11

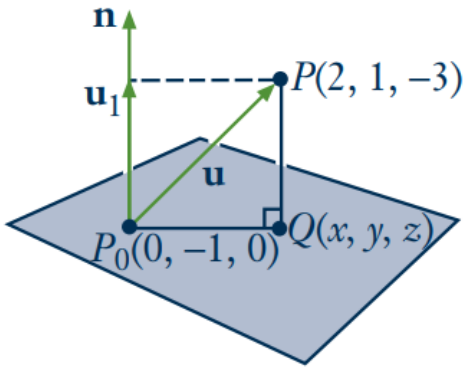


The plane in question has normal $n = \begin{bmatrix} 3 & -1 & 4 \end{bmatrix}^T$. Choose any point P_0 on the plane—say $P_0(0,-1,0)$ - and let $Q(x,y,z)$ be the point on the plane closest to P. The vector from P_0 to P is $u = \begin{bmatrix} 2 & 2 & -3 \end{bmatrix}^T$. Now erect **n** with its tail at P_0. Then $\overrightarrow{QP} = u_1$ and $\mathbf{u}_1$ is the projection of **u** on **n:**

$$u_1 = \frac{n \cdot u}{\|n\|^2} n = \frac{-8}{26} \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \frac{-4}{13} \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}$$

Solution 11

Hence the distance is $\left\|\overrightarrow{QP}\right\| = \|u_1\| = \dfrac{4\sqrt{26}}{13}$. To calculate the point Q, let



$$q = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } p_0 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

be the vectors of Q and $P0$. Then

$$q = p_0 + u - u_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ -3 \end{bmatrix} + \frac{4}{13}\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \dfrac{38}{13} \\ \dfrac{9}{13} \\ \dfrac{-23}{13} \end{bmatrix}$$

This gives the coordinates of Q(38/13, 13/9, -13/23).

The Cross Product

Definition 4.8

Given vectors $v_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, define the **cross product** $\mathbf{v}_1 \times \mathbf{v}_2$ by

$$v_1 \times v_2 = \begin{bmatrix} y_1 z_2 - z_1 y_2 \\ z_1 x_2 - x_1 z_2 \\ x_1 y_2 - y_1 x_2 \end{bmatrix}$$

Determinant Form of the Cross Product

If $v_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ are two vectors then

$$v_1 \times v_2 = \det \begin{bmatrix} i & x_1 & x_2 \\ j & y_1 & y_2 \\ k & z_1 & z_2 \end{bmatrix} = i \begin{vmatrix} y_1 & y_2 \\ z_1 & z_2 \end{vmatrix} - j \begin{vmatrix} x_1 & x_2 \\ z_1 & z_2 \end{vmatrix} + k \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}$$

where the determinant is expanded along the first column.

EXAMPLE 12

$$\text{Let } v_1 = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix}, \text{find } v_1 \times v_2.$$

Theorem 5

Let $\mathbf{v}$ *and* $\mathbf{w}$ *be vectors in* $\mathbb{R}^3$.

1. $\mathbf{v} \times \mathbf{w}$ *is a vector orthogonal to both* $\mathbf{v}$ *and* $\mathbf{w}$.

2. If $\mathbf{v}$ *and* $\mathbf{w}$ *are nonzero, then* $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ *if and only if* $\mathbf{v}$ *and* $\mathbf{w}$ *are parallel.*

EXAMPLE 13

Find the equation of the plane through $P(1, 3, -2)$, $Q(1, 1, 5)$, and $R(2, -2, 3)$.

4.3. More on the Cross Product

4.3. More on the Cross Product

Theorem 1

$$\text{If } u = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}, v = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \text{and w} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\text{then } u \cdot (v \times \text{w}) = \det \begin{bmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \\ z_0 & z_1 & z_2 \end{bmatrix}$$

4.3. More on the Cross Product

Theorem 2

Let $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *denote arbitrary vectors in* R^3.

1. $\mathbf{u} \times \mathbf{v}$ *is a vector.*
2. $\mathbf{u} \times \mathbf{v}$ *is orthogonal to both* $\mathbf{u}$ *and* $\mathbf{v}$.
3. $\mathbf{u} \times \mathbf{0} = \mathbf{0} = \mathbf{0} \times \mathbf{u}$.
4. $\mathbf{u} \times \mathbf{u} = \mathbf{0}$.
5. $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.
6. $(k\mathbf{u}) \times \mathbf{v} = k(\mathbf{u} \times \mathbf{v}) = \mathbf{u} \times (k\mathbf{v})$ *for any scalar* k.
7. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$.
8. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = (\mathbf{v} \times \mathbf{u}) + (\mathbf{w} \times \mathbf{u})$

4.3. More on the Cross Product

Theorem 3

Lagrange Identity

If **u** *and* **v** *are any two vectors in* R^3, *then*

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u.v)^2$$

Theorem 4

If **u** *and* **v** *are two nonzero vectors and* θ *is the angle between* **u** *and* **v**, *then*

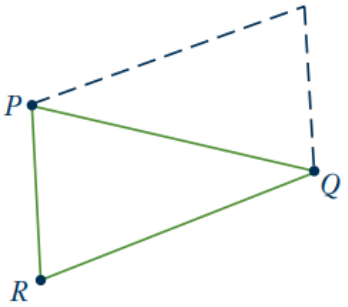
1. $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta$ = *area of the parallelogram determined by* **u** *and* **v**.

2. **u** *and* **v** *are parallel if and only if* $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.

4.3. More on the Cross Product

EXAMPLE 1

Find the area of the triangle with vertices $P(2, 1, 0)$, $Q(3, -1, 1)$, and $R(1, 0, 1)$.



4.3. More on the Cross Product

Theorem 5

The volume of the parallelepiped determined by three vectors $\mathbf{w}$, $\mathbf{u}$, *and* $\mathbf{v}$ *(Figure 2) is given by* $|\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})|$.

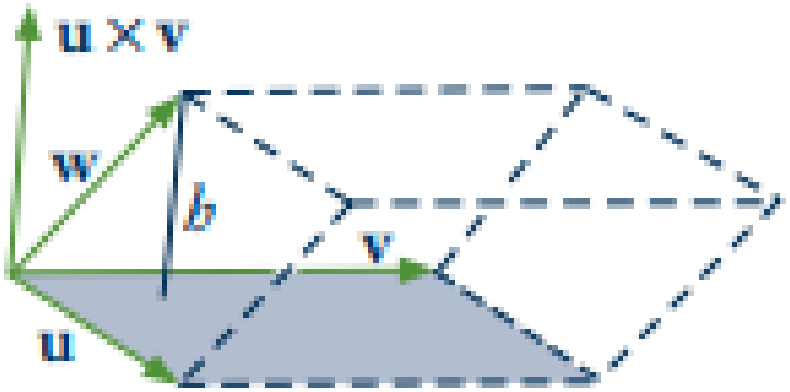


FIGURE 2

4.3. More on the Cross Product

EXAMPLE 2

Find the volume of the parallelepiped determined by the vectors

$$w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

4.4. Linear Operators on R^3

4.4. Linear Operators on R^3

Definition 4.9

A linear transformation

$$T : \mathbb{R}^n \to \mathbb{R}^n$$

is called a **linear operator** *on* R^n.

4.4. Linear Operators on R^3

$T:\mathbb{R}^n \to \mathbb{R}^n$ is said to be **distance preserving** if the distance between $T(\mathbf{v})$ and $T(\mathbf{w})$ is the same as the distance between $\mathbf{v}$ and $\mathbf{w}$ for all $\mathbf{v}$ and $\mathbf{w}$ in R^3; that is,

$$\|T(v)-T(\mathrm{w})\| = \|v-\mathrm{w}\| \text{ for all } v \text{ and } \mathrm{w} \text{ in } \mathbb{R}^3$$

Clearly reflections and rotations are distance preserving, and both carry $\mathbf{0}$ to $\mathbf{0}$, so the following theorem shows that they are both linear.

4.4. Linear Operators on R^3

Theorem 1

If $T:\mathbb{R}^3 \to \mathbb{R}^3$ *is distance preserving, and if* $T(\mathbf{0}) = \mathbf{0}$, *then* T *is linear.*

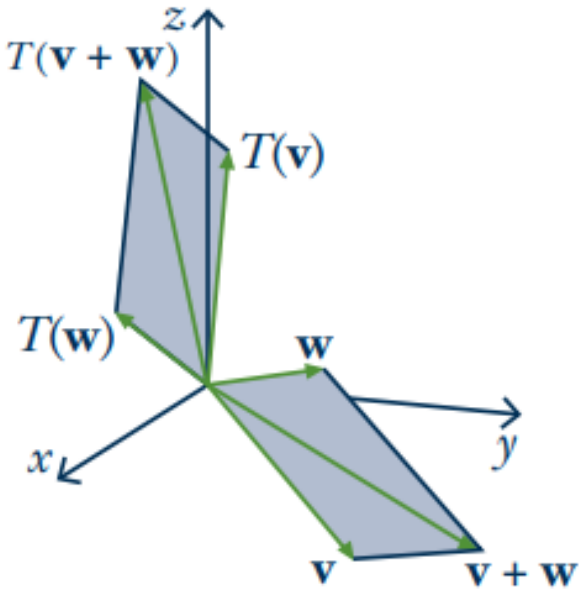


FIGURE 1

Reflections and Projections

We studied the reflection Q_m: $R^2 \rightarrow R^2$ in the line $y = mx$ and projection P_m: $R^2 \rightarrow R^2$ on the same line. We found that they are both linear and

$$Q_m \text{ has matrix } \frac{1}{1+m^2}\begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$$

$$\text{and } P_m \text{ has matrix } \frac{1}{1+m^2}\begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$$

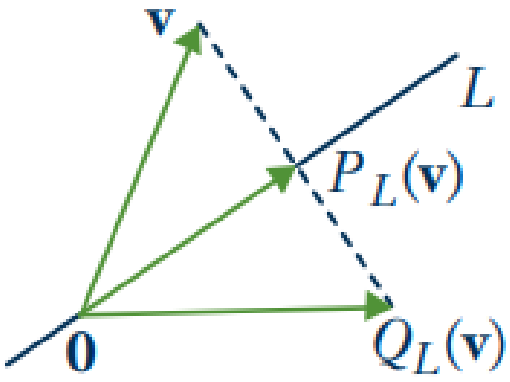


FIGURE 2

Reflections and Projections

EXAMPLE 1

In each case show that that T is either projection on a line, reflection in a line, or rotation through an angle, and find the line or angle.

(a) $T\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5}\begin{bmatrix} x+2y \\ 2x+4y \end{bmatrix}$

(b) $T\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{2}\begin{bmatrix} x-y \\ y-x \end{bmatrix}$

(c) $T\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} -x-y \\ x-y \end{bmatrix}$

(d) $T\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5}\begin{bmatrix} -3x+4y \\ 4x+3y \end{bmatrix}$

(e) $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ -x \end{bmatrix}$

(f) $T\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{2}\begin{bmatrix} x-\sqrt{3}y \\ \sqrt{3}x+y \end{bmatrix}$

Reflections and Projections

Solution 1

(a) projection on the line $y = 2x$.

(b) projection on the line $y = -x$.

(c) rotation through the angle $\frac{3\pi}{4}$ or $135°$.

(d) reflection in the line $y = 2x$.

(e) reflection in the line $y = -x$.

(f) rotation through the angle $\frac{\pi}{3}$ or $60°$.

Reflections and Projections

Theorem 2

Let L denote the line through the origin in R^3 with direction vector $d = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$. *Then P_L and Q_L are both linear and*

P_L *has matrix* $\dfrac{1}{a^2+b^2+c^2}\begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$,

Q_L *has matrix* $\dfrac{1}{a^2+b^2+c^2}\begin{bmatrix} a^2-b^2-c^2 & 2ab & 2ac \\ 2ab & b^2-a^2-c^2 & 2bc \\ 2ac & 2bc & c^2-a^2-b^2 \end{bmatrix}$.

Reflections and Projections

EXAMPLE 2

Find the reflection of $v = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ in the line with equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\frac{1}{a^2+b^2+c^2} \begin{bmatrix} a^2-b^2-c^2 & 2ab & 2ac \\ 2ab & b^2-a^2-c^2 & 2bc \\ 2ac & 2bc & c^2-a^2-b^2 \end{bmatrix}.$$

Reflections and Projections

Theorem 3

Let M denote the plane through the origin in R^3 with normal

$n = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$. *Then P_M and Q_M are both linear and*

P_M has matrix $$\frac{1}{a^2+b^2+c^2}\begin{bmatrix} b^2+c^2 & -ab & -ac \\ -ab & a^2+c^2 & -bc \\ -ac & -bc & a^2+b^2 \end{bmatrix},$$

Q_M has matrix $$\frac{1}{a^2+b^2+c^2}\begin{bmatrix} b^2+c^2-a^2 & -2ab & -2ac \\ -2ab & a^2+c^2-b^2 & -2bc \\ -2ac & -2bc & a^2+b^2-c^2 \end{bmatrix}.$$

Reflections and Projections

EXAMPLE 3

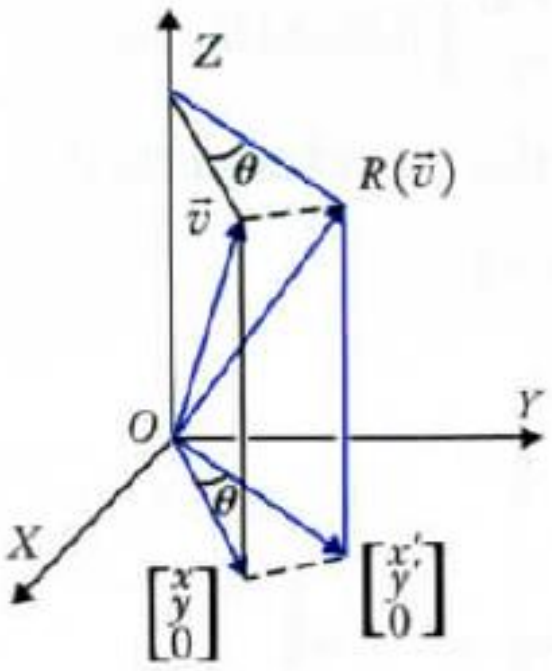
Find the projection of $v = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ in the plane with equation

$$3x - 5y + 2z = 0$$

$$\frac{1}{a^2 + b^2 + c^2} \begin{bmatrix} b^2 + c^2 & -ab & -ac \\ -ab & a^2 + c^2 & -bc \\ -ac & -bc & a^2 + b^2 \end{bmatrix},$$

Rotation

- The rotation of R^3 about the Z axis through an angle θ from the X axis to Y axis



R_θ *has matrix*

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

EXAMPLE 4

Find the rotation of $v = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ about the z axis through $\theta = \frac{\pi}{4}$

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transformations of Areas and Volumes

Theorem 4

If $T:\mathbb{R}^3 \to \mathbb{R}^3$ (*or* $\mathbb{R}^2 \to \mathbb{R}^2$) *is a linear operator, the image of the parallelogram determined by vectors* $\mathbf{v}$ *and* $\mathbf{w}$ *is the parallelogram determined by* $T(\mathbf{v})$ *and* $T(\mathbf{w})$.

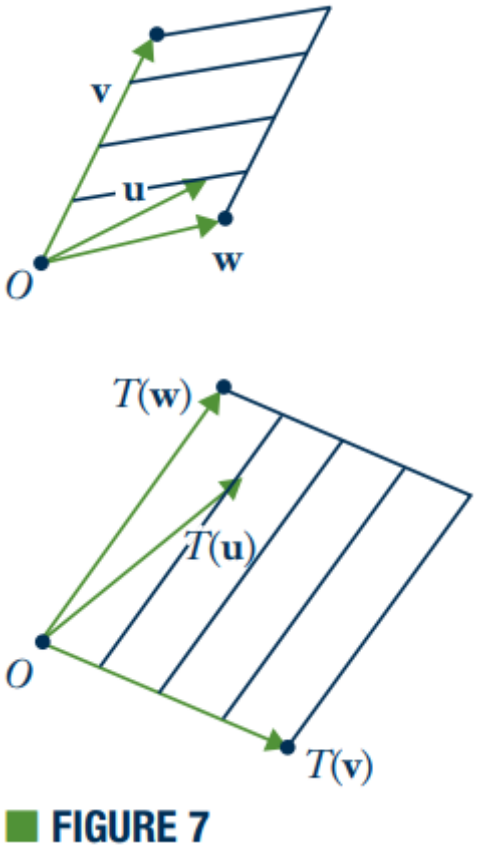


FIGURE 7

Transformations of Areas and Volumes

Theorem 5

Let vol(**u**, **v**, **w**) *denote the volume of the parallelepiped determined by three vectors* **u**, **v**, *and* **w** *in* R^3, *and let* area(**p**, **q**) *denote the area of the parallelogram determined by two vectors* **p** *and* **q** *in* R^2. *Then*:

1. *If A is a* 3×3 *matrix, then* vol(A**u**, A**v**, A**w**) = |det(A)| · vol(**u**, **v**, **w**).
2. *If A is a* 2×2 *matrix, then* area(A**p**, A**q**) = |det(A)| · area(**p**, **q**).

4.3. More on the Cross Product

EXAMPLE 2

Find the volume of the parallelepiped determined by the vectors Aw, Au, Av with

$$w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 2 & 1 & -1 \end{bmatrix}$$

4.3. More on the Cross Product

Theorem 5

The volume of the parallelepiped determined by three vectors $\mathbf{w}$, $\mathbf{u}$, *and* $\mathbf{v}$ *(Figure 2) is given by* $|\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})|$.

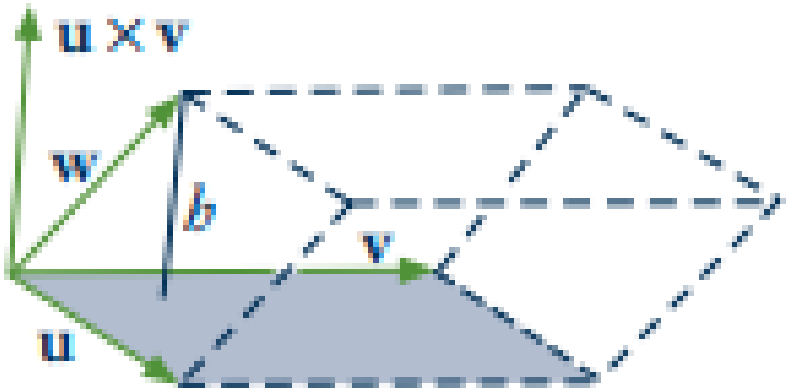


FIGURE 2

4.5. An Application to Computer Graphics

4.5. An Application to Computer Graphics

Consider displaying the letter *A*. In reality, it is depicted on the screen, as in Figure 1, by specifying the coordinates of the 11 corners and filling in the interior. For simplicity, we will disregard the thickness of the letter, so we require only five coordinates as in Figure 2. This simplified letter can then be stored as a data matrix.

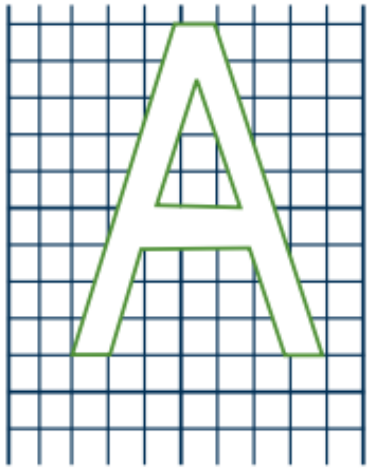


FIGURE 1

$$\begin{matrix} \textit{Vertex} & 1 & 2 & 3 & 4 & 5 \end{matrix}$$

$$D = \begin{bmatrix} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{bmatrix}$$

where the columns are the coordinates of the vertices in order

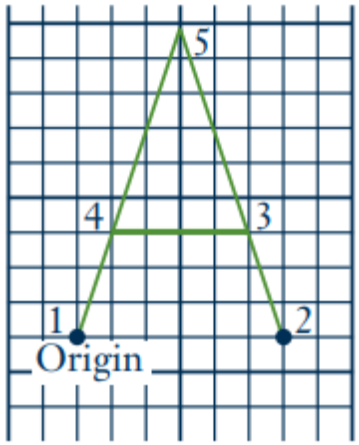


FIGURE 2

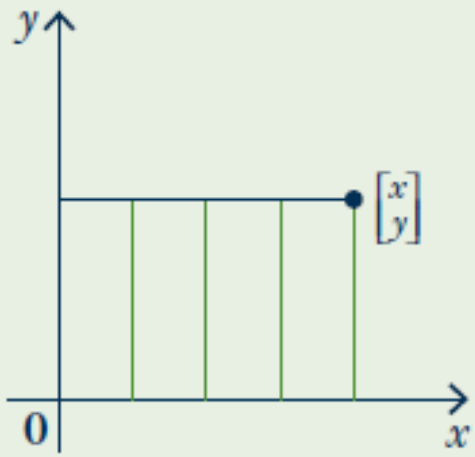
Example

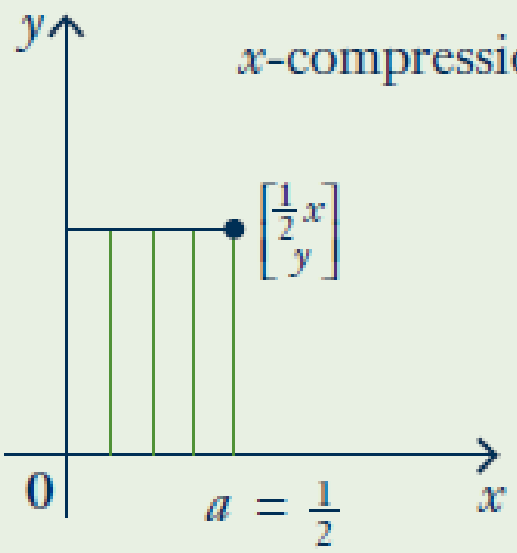
$$A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix} \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}$$

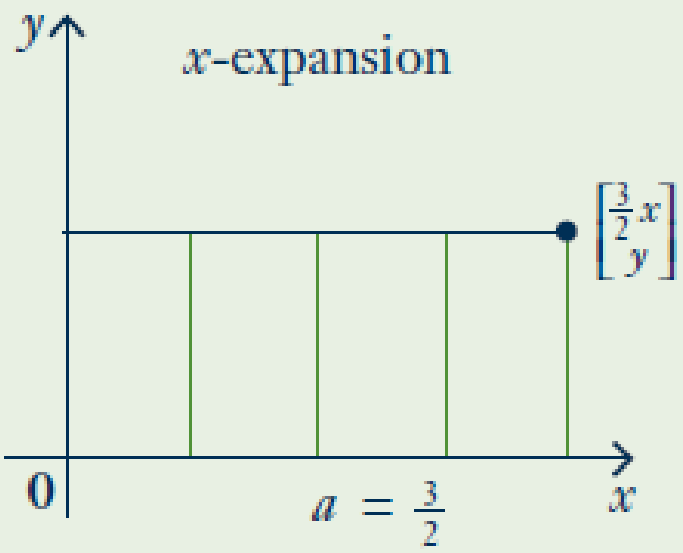
$$\begin{bmatrix} x \\ y \end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ y \end{bmatrix}$$

- a>1:**X-expansion** of R^2
- 0<a<1:**X-compression** of R^2

Example







Example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix} \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ by \end{bmatrix}$$

- b>1:**Y-expansion** of R^2
- 0<b<1:Y-**compression** of R^2

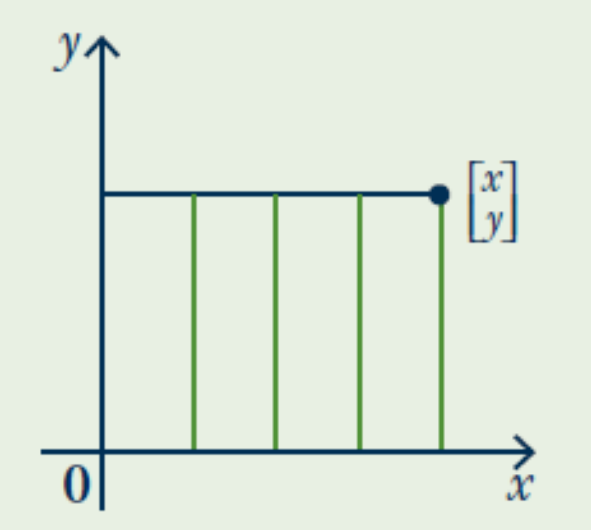
Example

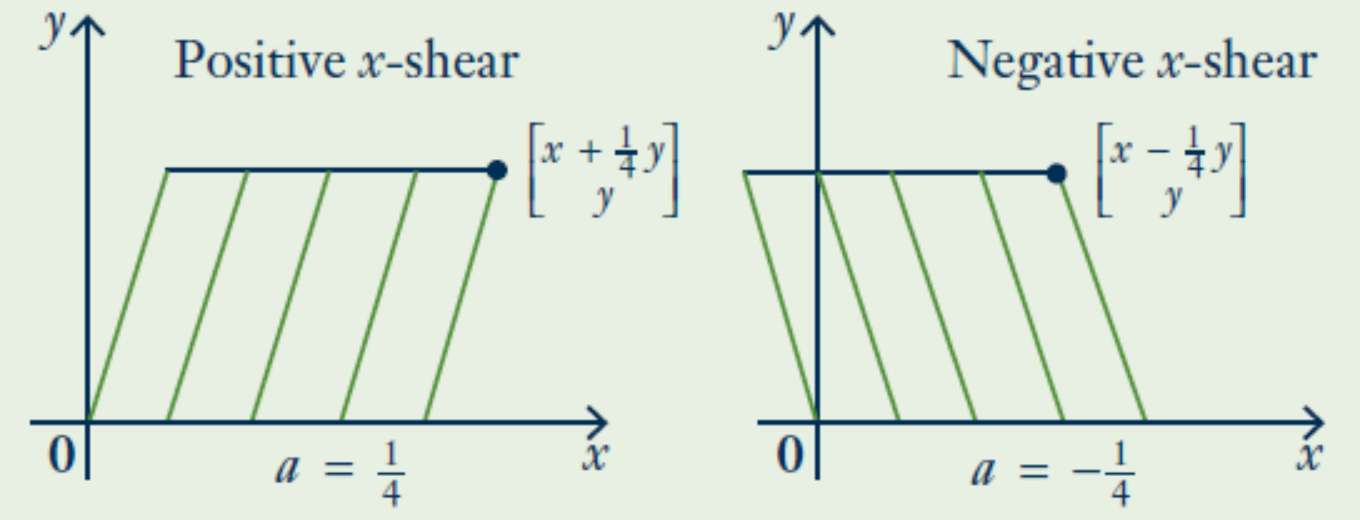
$$A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}$$

- a>0: Positive **X-shear** of R^2
- a<0: Negative **X-shear** of R^2

$$\begin{bmatrix} x \\ y \end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+ay \\ y \end{bmatrix}$$

Example


y
x
y
0
x


y
Positive x-shear
x + 1/4 y
y
0
a = 1/4
x
y
Negative x-shear
x − 1/4 y
y
0
a = −1/4
x

4.5. An Application to Computer Graphics

If we want to transform the letter by a 2 × 2 matrix *A*, we left-multiply this data matrix by *A* (the effect is to multiply each column by *A* and so transform each vertex).

For example, we can slant the letter to the right by multiplying by an *x*-shear matrix

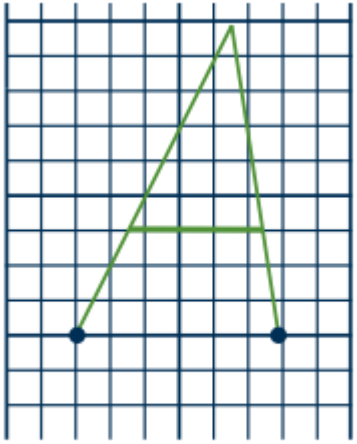


FIGURE 3

$$A = \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

The result is the letter with data matrix

$$A = \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 5.6 & 1.6 & 4.8 \\ 0 & 0 & 3 & 3 & 9 \end{bmatrix}$$

4.5. An Application to Computer Graphics

If we want to make this slanted matrix narrower, we can now apply an *x*-scale matrix

$$B = \begin{bmatrix} 0.8 & 0 \\ 0 & 1 \end{bmatrix}$$

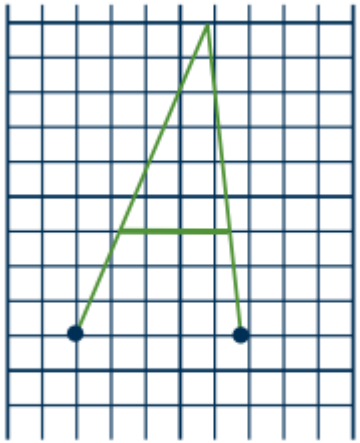


FIGURE 4

that shrinks the *x*-coordinate by 0.8. The result is the composite transformation

$$BAD = \begin{bmatrix} 0 & 4.8 & 4.48 & 1.28 & 3.84 \\ 0 & 0 & 3 & 3 & 9 \end{bmatrix}$$

which is drawn in Figure 4.

4.5. An Application to Computer Graphics

On the other hand, we can rotate the letter about the origin through 30° by multiplying by the matrix

$$R_{\frac{\pi}{6}} = \begin{bmatrix} \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} \\ \sin\frac{\pi}{6} & \cos\frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}$$

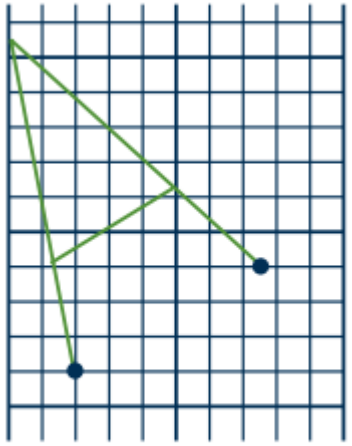


FIGURE 5

This gives

$$R_{\frac{\pi}{6}} D = \begin{bmatrix} 0 & 5.196 & 2.83 & -0.634 & -1.902 \\ 0 & 3 & 5.098 & 3.098 & 9.294 \end{bmatrix}$$